2026 Hacker Badge — Factory Flash & Functional Test Procedure

Production test instructions for assembly / fabrication

Revision A

1. Purpose & Scope

This document describes how to **flash firmware** onto each assembled 2026 Hacker Badge and how to **functionally test** every board before it ships. The whole test is driven from a web browser and the badge's own built-in factory test — no special programmer, jig, or command-line tools are required.

Target board: 2026 Hacker Badge **Processor:** Espressif ESP32-C3 (RISC-V, USB-Serial-JTAG built in) **Connection:** USB-C cable, direct to a computer (no external adapter needed)

Quantity for this run: 50 boards. Every board must pass. Estimated time once the operator is trained: **about 30 seconds per board.**

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2. Equipment Required (per test station)

Item	Notes
Computer	Windows, macOS, or Linux
Web browser	**Desktop Google Chrome, Microsoft Edge, or Opera.** Firefox and Safari are **not** supported.
USB-C cable	Data-capable (not charge-only).
Internet **or** local copy of the flashing page	See Section 3.
2 × AA batteries	One set only — **reuse the same pair for every board** (Section 6).
NFC card / tag	**Optional.** Any 13.56 MHz card (transit card, hotel key, blank tag). Only needed if you want to test a card scan; see Section 5, Screen 3.

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3. One-Time Station Setup

1. Open the flashing web page in Chrome/Edge/Opera:

   `https://hackathon.github.io/badge-firmware/`

   *(If the station has no reliable internet, the page can also be run from a local copy — request the offline package and open `index.html` through a local web server. The page must be served over **HTTPS or localhost**; opening the file directly will not allow USB access.)*

2. Confirm the page loads and shows the **"Flash a Badge"** button. If instead it shows a red "this browser can't access serial devices" message, you are in the wrong browser — switch to Chrome, Edge, or Opera.

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4. Flashing Procedure (per board)

Perform this for **every** board.

1. **Connect** the badge to the computer with the USB-C cable.
2. Click **"Flash a Badge"** on the web page.
3. A small window lists serial ports. Select the badge's port (typically shown as *"USB JTAG/serial debug unit"* or similar) and click **Connect**.
4. Flashing begins **automatically** — do not touch the cable. The progress bar fills and the status shows *Connecting → Loading firmware → Flashing → Restarting*.
5. Wait for the green **"Flashed successfully"** success screen.
6. The badge **automatically reboots into the built-in Factory Test** (Section 5). The first screen is the **Buttons** test. **If it does not reboot on its own within a few seconds, unplug the USB-C cable and plug it back in.**

If flashing fails (red error screen): unplug, re-seat the cable, and click **"Try Again."** If it still fails, try a different USB port and a different USB port — most failures are cable or port problems.

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5. Functional Test Procedure (per board)

After flashing, the badge boots directly into the **Factory Test**. The operator interacts with the **badge's own buttons and screen** — not the computer. The test is bilingual (English + Chinese) on screen.

At the top of every screen is a colored status strip with three cells — **BTN, LED, NFC** — that turn **green** as each stage passes. The current stage has a cyan outline.

Screen 1 — Buttons

- Title shows **"Buttons"** / 按下所有按钮 ("Press every button").
- A pad is shown on screen for each button. **Press each of these 8 buttons once: Up, Down, Left, Right, A, B, Home, Start.**
- Each pad turns **green** when its button registers. When all 8 are green the test **advances on its own** to the LED screen.
- A button that never turns its pad green, or a pad that turns green without being pressed (stuck button), means this board fails the button test.

Screen 2 — LEDs

- Title shows **"LEDs"**. On-screen prompt: **"A = working B = broken"** / 灯光正常按 *A* 故障按 *B*.
- The LED strip cycles **Red → Green → Blue → White** (the current color name is shown on screen). Let it run through at least one full cycle.
- Watch all **6 LEDs**. If every LED lights and shows the correct colors together, press **A** (working). If any LED is dark or shows a wrong color, press **B** (broken).

Screen 3 — NFC (optional card scan)

- Title shows **"NFC"**. On-screen prompt: **"Scan a card, or A to skip"** / 扫描任意卡片或按 *A* 跳过.
- **If you have an NFC card on hand:** hold it flat against the back of the badge. When it reads, the test advances automatically and NFC is marked passed.
- **If you do not have a card:** press **A** to skip. This is fine — the **NFC reader chip is still verified automatically** in the background (reported as "NFC chip" on the Summary). Only the optional card-read is skipped.

Screen 4 — Summary

The Summary screen lists every check with a status mark:

- **Green check** = passed

- **Red X** = failed
- **Grey dash** = skipped / not run (e.g. NFC card scan when skipped)

The five lines are: **Buttons, LEDs, NFC, Accel, NFC chip.**

A board PASSES when these four show a green check: Buttons, LEDs, Accel, and NFC chip. The **NFC** line may be a green check (card scanned) **or** a grey dash (card skipped) — both are acceptable.

Press **A** on the Summary to run the test again if you need to recheck anything.

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6. Battery Test (2 × AA)

This confirms the board runs on battery power, not just USB. **Use one set of AA batteries and reuse it for every board.**

1. After the board passes the USB functional test, **unplug the USB-C cable.**
2. Insert the **2 × AA batteries.** Match the flat **(−)** end and the bump **(+)** end of each battery to the **−** and **+** marks printed in the battery holder.
3. **Turn ON the power switch** at the top edge of the badge.
4. Confirm the **screen powers on** and the badge starts the factory test on battery power alone.
5. **Turn OFF the power switch, remove the batteries,** and keep them for the next board.

PASS: the screen lights up and runs on battery power. **FAIL:** screen stays dark, flickers, or will not start on batteries.

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7. Important Notes

- **Do not write on, label, sticker, or otherwise mark the badges.** Keep passed and failed boards in separate trays/areas instead. The badges ship as-is and any marking is visible to the end user.
- **Reuse the same AA batteries** for the battery test on every board — they are only inserted briefly to confirm power.
- A board only needs to be flashed once. If a board has already passed and you re-flash it, that is harmless.